ConceptAlign: A Human-in-the-Loop Large Language Model Framework for Concept Generation and Alignment

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Abstract

This work studies Concept Bottleneck Models (CBM), which can explain predictions of deep neural networks (DNNs) through human-understandable concepts. A key challenge in applying CBMs is their reliance on extensive concept labels manually annotated by domain experts. While recent studies adopt large language models (LLMs) to automatically generate concepts, these methods typically generate them based on known class labels rather than directly from unlabeled input data. To address this limitation, we propose ***ConceptAlign****, a human-in-the-loop LLM framework for concept generation and alignment through active learning. The key challenge is to minimize labeling effort by identifying most informative data–attribute pairs, which requires reliable uncertainty estimation during both the cold-start and fine-tuning phases. To this end, we develop a hybrid uncertainty estimation method that combines (1) prompt-ensemble uncertainty, measured via prompt-ensemble variance over diverse concept exemplars during the cold start phase, and (2) LoRA dropout uncertainty, obtained through Monte Carlo dropout during LoRA fine-tuning. Extensive experiments on multiple benchmarks demonstrate that ConceptAlign extracts more accurate and human-aligned concepts than the state-of-the-art methods using at most 300 labeled samples. Moreover, it increases classification accuracy by 2% and 5% on bird and medical image classification tasks, respectively, while also providing more accurate interpretability.*

1. Introduction

Concept-based learning [6, 8] is an emerging interpretable paradigm that aims to explain deep neural network (DNN) predictions through human-understandable concepts. A representative example is the Concept Bottleneck Models (CBMs), which first learn a set of textual concept bottlenecks and then predict the target class based on these learned concepts. The key advantage of CBMs lies in their ability to integrate intrinsic interpretability directly into the model architecture, thereby enabling scalable and transparent explanations of model predictions [26].

Earlier works [13, 19, 30] have employed CBMs to explain predictions of classical DNNs in the image domain. However, a major limitation of these approaches is their reliance on large sets of concept labels manually annotated by domain experts. To overcome this problem, recent studies [15, 26, 35] have adopted LLMs to generate interpretable concepts based on **class labels**. Nonetheless, this line of approaches has a major limitation: the extracted concepts are identical for all samples within the same class, even when individual instances differ in their concept values. For example, as shown in Figure 2, birds from the same category, such as European Goldfinches, may exhibit different face and head colors. However, the concepts generated by baseline methods (highlighted in blue) remain identical for these birds. In practice, obtaining class labels is often challenging in high-stakes domains such as medical imaging. In contrast, humans naturally identify interpretable concepts, such as shape, size, and color, directly from observations, without relying on predefined labels.

Motivated by this, we propose **ConceptAlign**, a human-in-the-loop LLM framework that extracts fine-grained concepts directly from observed data via active learning. *The key challenge lies in minimizing the number of both labeled concepts and data samples by human graders during both the cold-start and fine-tuning phases*. To address this challenge, we introduce a hybrid uncertainty method that combines prompt-ensemble uncertainty and dropout uncertainty to select and annotate the most informative data-attribute pairs. Specifically, we first estimate the epistemic uncertainty of generated concepts during the cold-start phase by by measuring the variability induced by different in-context examples. However, the LLM gradually becomes less sensitive to changes in the prompt, which makes uncertainty estimation increasingly difficult. Therefore, we employ Monte Carlo dropout to quantify the uncertainty of concept generation during LoRA [12] fine-tuning phase. This hybrid approach significantly reduces the amount of human effort

	European Goldfinch	Florida Jay	Rose Breasted Grosbeak	
	ConceptAlign (Ours)	**LF-CBM**	**VLG-CBM**	**Hybrid-CBM**
①	1. small and plump body 2. short, thick and conical beak 3. short and notched tail 4. pale brown beak 5. pale brown face 6. pale brown chest 7. black wings with yellow patches 8. pinkish brown feet	1. a black cap on the head 2. a long, slender beak 3. a pointed beak 4. a small, delicate body 5. a small, forked tail 6. a small, round body 7. a yellow breast 8. black and white wings	1. small forked tail 2. long slender beak 3. pointed beak 4. yellow breast 5. black cap on the head 6. black and white wings	1. often using thistles for nesting material 2. seen in pet stores 3. hatch in about two weeks 4. small, sprightly finch with a bright yellow body and black wings 5. population that is increasing across North America 6. common bird in the eastern United States and Canada 7. small bird with yellow eyes and yellow plumage 8. common bird in the wild and its yellow plumage
②	1. clawed feet 2. grayish brown crown 3. grayish black beak 4. grayish brown face 5. grayish brown throat 6. white belly 7. blue wings with black tips 8. black feet	1. long, black beak 2. black head 3. blue body 4. blue, gray back 5. blue wings 6. white chest and belly	1. a black head 2. a blue body 3. a blue-gray back 4. a long, black beak 5. a small body 6. a white chest 7. a white chest and belly 8. blue wings	1. found in woodlands and forest throughout Florida 2. nests in tree cavities or in man-made nest boxes 3. blue legs and feet 4. eat insects, berries and acorns 5. black-and-white striped tail 6. a blue and white bird with a blue and white neck 7. the blue bird is in the southern United States 8. a blue and white bird with a blue and yellow eyes
③	1. small and compact body 2. short, thick and conical beak 3. pale pinkish beak 4. black face 5. black throat 6. white chest with red patches 7. white belly 8. black wings with white bars	1. a black head 2. a black tail with white tips 3. a large, red breast 4. a long, black beak 5. a white belly 6. white stripes on the wings 7. yellow wings	1. long black beak 2. black head 3. large red breast 4. white stripes on the wings 5. yellow wings 6. black tail with white tips 7. white belly	1. wingspan is around 30 cm 2. bird's nest is cup-shaped structure made of twigs, leaves 3. stout, cone-shaped bills that are pinkish-gray in color 4. form small flocks with other grosbeaks 5. lifespan of up to 15 years in the wild 6. red bird with black and white feathers on the side of road 7. red and black bird with red and white eyes 8. red and white bird with black and white feather

Figure 1. Case studies of the proposed ConceptAlign model and baseline methods. The **black text** indicates correctly generated concepts; the **red text** denotes incorrectly generated concepts; and the **blue text** represents generated concepts that are identical across all birds within the same category, even if individual birds possess some differing concepts. We can observe that our method can accurately generate fine-grained concepts while baselines will generate inaccurate concepts.

required for data annotation.

We evaluate the proposed ConceptAlign framework on two benchmark datasets commonly used for CBMs. Extensive experimental results demonstrate that our method identifies concepts that are both more accurate and better aligned with human understanding compared to the competitive baselines. Moreover, the extracted concepts enhance both classification accuracy (improved by 3% on average) and interpretability in downstream tasks.

The main contributions of this work are summarized as follows:

- We develop ConceptAlign, a novel human-in-the-loop LLM framework that effectively generates accurate and human-aligned concepts through active learning.
- We propose a hybrid uncertainty estimation method that integrates prompt-ensemble and dropout uncertainties to select and annotate the most informative data–attribute pairs, thereby reducing human annotation effort.
- Extensive evaluations on two benchmark datasets demonstrate that our approach outperforms competitive baselines in both concept generation accuracy and downstream classification performance.

2. Related Works

Concept Bottleneck Models. Concept Bottleneck Models (CBMs) [13] aim to provide explicit interpretability for DNNs by learning human-understandable concepts from observed data. Early studies [13, 20, 30, 34] have applied CBMs to explain image classification tasks in high-stakes domains such as medical imaging. However, a major limitation of these approaches is their reliance on domain experts to manually annotate concepts for all data samples. More recently, several works [18, 26, 27, 35] have extended CBMs to improve the interpretability of large language models (LLMs) in both text generation and image classification tasks. For example, the latest Hybrid-CBM framework addresses the issue of incomplete predefined concepts. Nevertheless, most existing methods still depend on known class labels to generate concepts, which demands substantial human effort and annotation cost. In practice, humans often recognize interpretable attributes, such as shape, size, and color, directly from the data rather than from class labels. In contrast to prior works, we aim to generate human-understandable concepts directly from observed data through active learning.

Uncertainty Estimation in LLMs. Quantifying the uncertainty of LLMs remains a challenging problem. There are two different types of uncertainty quantification methods: (i) aleatoric uncertainty, which is caused by inherent randomness or noise in the data; and (ii) epistemic uncertainty, which arises from lack of knowledge due to limited or unrepresentative data. Some studies [1, 37] have employed aleatoric uncertainty techniques like prompts to quantify the uncertainty of LLMs while others have adopted epistemic uncertainty like Monte Carlo dropout [4, 17]. In this work, we combine prompt ensembling uncertainty and Monte Carlo dropout to quantify the uncertainty of concept generation by LLMs. The proposed approach identifies the most informative data–concept pairs, which are then labeled by humans.

LLM-based Active Learning. Active Learning (AL) [28] aims to improve model efficiency and performance by selecting the most informative data points for labeling and training. Recent studies [32] have leveraged large language models (LLMs) for data selection and annotation to reduce human effort. The central idea of LLM-based active learning is to input unlabeled data into LLMs to identify informative samples based on uncertainty estimation, which are then annotated by either LLMs or humans. For instance, NoiseAL [36] integrates LLMs with small models (SMs) to identify promising unlabeled data and subsequently annotate them. Similarly, Wang et al. [31] introduced a human–LLM collaborative annotation framework, in which LLMs first generate annotations that are later verified by human experts. Unlike prior studies, this work integrates active learning into the supervised fine-tuning for concept alignment through a hybrid uncertainty approach.

3. Preliminaries

Problem statement. Suppose we have a set of N observed data samples, denoted as $\mathcal{I} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N\}$, where $\boldsymbol{x}_i$ represents the i-th data sample. Each sample is characterized by M high-level *attributes* (e.g., beak shape), and each attribute $m \in \{1, \cdots, M\}$ can take on one of V_m possible *concept values* (e.g., "short and thin beak", "long and thick beak"). The objective of this work is to generate fine-grained concepts for each data sample using LLMs. The key challenge lies in minimizing the annotation effort required for labeling both the concepts and the data samples.

Notations. We describe the main notations used in this work in Table 3 in Appendix A.

4. Proposed Method

In this section, we first introduce ConceptAlign, an LLM-based active learning framework for concept generation and alignment. We then summarize the algorithm used for uncertainty quantification in concept generation.

4.1. ConceptAlign

As mentioned above, the key challenge is to minimize the number of labeled concepts and data samples by human graders. To address this, we introduce ConceptAlign, a novel human-in-the-loop LLM framework that can generate human-understandable concepts through active learning. As shown in Figure 2, we propose a hybrid epistemic uncertainty estimation method that combines prompt-ensemble uncertainty and LoRA dropout uncertainty to identify most informative data-attribute pairs during both the cold-start and LoRA fine tuning phases. The core idea is that we first use prompt-ensemble uncertainty to identify samples whose concept values vary the most using different in-context examples. This method is simple and effective for initial supervision in the first cold-start phase. For the second phase, we adopt LoRA dropout to expose true epistemic uncertainty when the model is overfitted to a few labeled data, highlighting samples that will most effectively correct overfitting and further align concept values with human judgments.

Phase 1: Prompt-Ensemble Uncertainty. Inspired by recent uncertainty estimation methods for LLMs [11, 14], we estimate epistemic uncertainty of generated concepts during the cold-start phase by measuring the variability of concept predictions across in-context examples. This method is performed in the following **five steps**: 1) we input data samples along with varying in-context prompts to generate multiple concept values for each attribute; 2) we utilize natural language inference (NLI) to cluster the generated concept values [14], and derive a categorical distribution over the concept clusters; 3) we calculate the prompt-ensemble uncertainty as the entropy of this distribution; 4) we weight the uncertainty score based on semantic similarity to discount uncertainty arising from highly similar concept clusters; 5) we rank all data samples by their corresponding uncertainty scores to select a training set to fine-tune the LLM.

Formally, for each input data sample $\boldsymbol{x}$, we fix the task instruction and the attribute template in its prompt $\boldsymbol{q}$, and insert K distinct sets of in-context examples $\{\mathcal{E}^k\}_{k=1}^K$ sampled from a few initial concept annotations (see Fig. 3). For each attribute m, we input a pair of image and the k-th in-context prompt into the LLM to obtain K free-form concept predictions $\{v_m^k\}_{k=1}^K$, where v_m^k indicates the k-th textual candidate lying in the concept value space V_m for attribute m. To obtain a categorical distribution over the concept values $\{v_m^k\}_{k=1}^K$, we apply NLI-based clustering [14] to group the concept values for each attribute into semantically coherent clusters. Specifically, for each attribute m and its set of generated concepts $\{v_m^k\}_{k=1}^K$, we run an NLI model [10] on every pair (v_m^i, v_m^j). We say that two concept values v_m^i and v_m^j are NLI-equivalent if:

$$\hat{y}_{\text{NLI}}(v_m^i \Rightarrow v_m^j) = \hat{y}_{\text{NLI}}(v_m^j \Rightarrow v_m^i) = \text{entail}$$

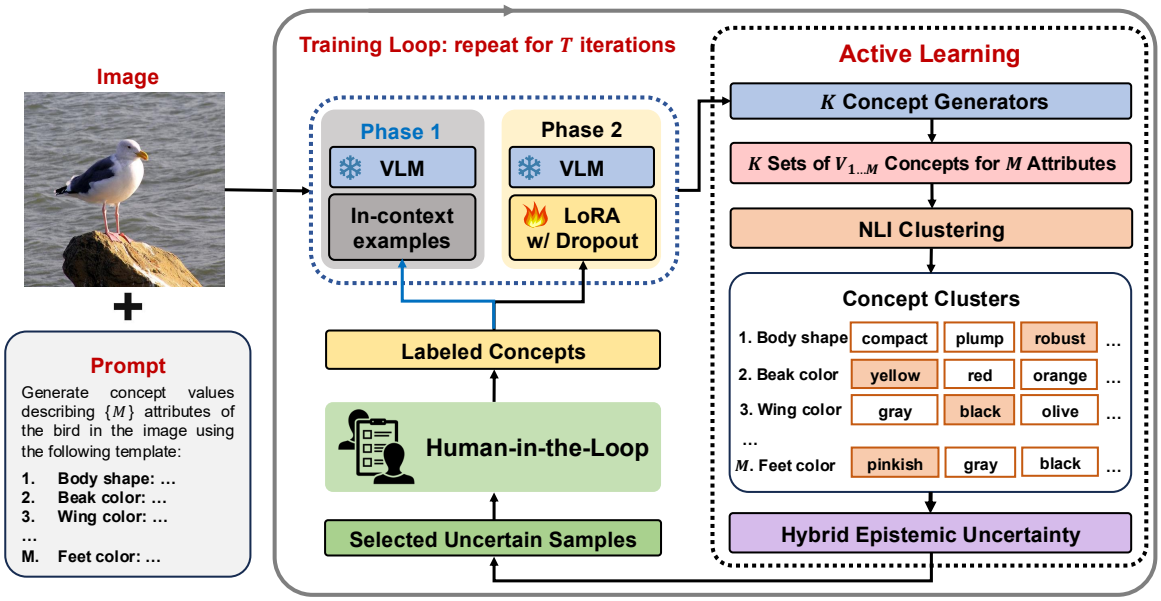

Figure 2. The overall framework of ConceptAlign. It adopts a hybrid uncertainty estimation method that combines prompt-ensemble uncertainty and dropout uncertainty to identify the most informative data-attribute pairs for fine-tuning LLMs.

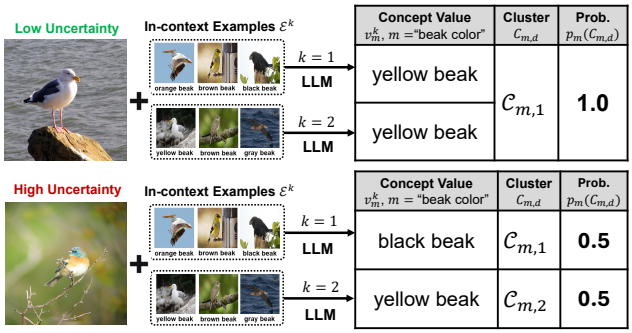

Figure 3. A toy example of the prompt-ensemble uncertainty estimation procedure. With a fixed prompt, we pass one image through the LLM twice ($K = 2$), each time with a different set of 3 in-context examples. Then we use NLI-clustering to group the generated concept values into semantically coherent clusters. From the figure, if the model is uncertain, the two passes will result in different clusters; if confident, they concentrate in one. Consequently, we calculate the cluster probabilities by normalizing their counts over two passes, then define prompt-ensemble uncertainty as the entropy of this probability distribution.

where $\hat{y}_{\text{NLI}}(v_m^i \Rightarrow v_m^j) \in \{\text{contradict}, \text{neutral}, \text{entail}\}$ denotes the label predicted by the NLI model when treating v_m^i as the premise and v_m^j as the hypothesis. We then construct an undirected graph whose nodes are $\{v_m^k\}_{k=1}^K$ and place an edge between v_m^i and v_m^j whenever they are NLI-equivalent. The connected components of this graph define D_m semantic clusters $\{\mathcal{C}_d\}_{d=1}^{D_m}$ for each attribute m, with each $\mathcal{C}_d \subseteq \{v_m^{(k)}\}_{k=1}^K$ containing mutually equivalent concept values. Then, we obtain a categorical distribution over the concept clusters by normalizing their counts over K passes, which is given by:

$$p_m(\mathcal{C}_{m,d}) = \frac{|\mathcal{C}_{m,d}|}{K}, \tag{1}$$

where $|\mathcal{C}_{m,d}| = \sum_{k=1}^K \mathbf{1}\{v_m^k \in \mathcal{C}_{m,d}\}$ is the number of concept values assigned to cluster $\mathcal{C}_{m,d}$. Afterwards, we calculate the entropy of the categorical distribution over D_m concept clusters to quantify the model's uncertainty for attribute m as follows:

$$U_m^{prompt} = H(p_m) = -\sum_{d=1}^{D_m} p_m(\mathcal{C}_{m,d}) \log p_m(\mathcal{C}_{m,d}) \tag{2}$$

While NLI-based clustering can effectively groups paraphrases of candidate concept values for each attribute into clusters, one remaining issue is that different clusters can still be very close in meaning (e.g., "short and thin beak" vs. "medium and thin beak"). Additionally, the entropy scores $H(p_m)$ treats such small differences the same as disagreements between truly distinct clusters for all attributes. To prioritize attributes where uncertainty reflects substantive semantic disagreement, we make each $H(p_m)$ distance-aware by weighting entropy with an inter-cluster dissimilarity term. Given the concept clusters $\{\mathcal{C}_{m,d}\}_{d=1}^{D_m}$ for each attribute m, we select a textual prototype $\mu_{m,d}$ for each cluster $\mathcal{C}_{m,d}$ whose average pairwise NLI equivalence probability to other members is maximal. To incorporate semantic distance into the uncertainty score, we compute embedding-based dissimilarity between cluster prototypes. Specifically, let $\boldsymbol{S} \in [-1,1]^{D_m \times D_m}$ denote the cosine similarity matrix between cluster prototypes $\{\mu_{m,d}\}_{d=1}^{D_m}$, obtained from a

sentence embedding model [24]. To avoid weighting highly dissimilar clusters, we adjust the cosine similariy matrix $\boldsymbol{S}$ between cluster prototypes using a high baseline s_{base} and normalize the results to $[0, 1]$ element-wise as follows:

$$\boldsymbol{S}_{ij}^{\text{adj}} = \max\left(0, \frac{\boldsymbol{S}_{ij} - s_{\text{base}}}{1 - s_{\text{base}}}\right) \tag{3}$$

where i, j index the rows and columns of the similarity matrix $\boldsymbol{S}$, corresponding to cluster prototypes for attribute m.

Inspired by Rao [23], given the concept cluster distribution p_m for each attribute m (Eq. 1), we then compute a concept concentration score as:

$$q_m = p_m^\top \cdot \boldsymbol{S}^{\text{adj}} \cdot p_m \in [0, 1] \tag{4}$$

Here, it can be observed that when the concept clusters are close in the embedding space, the concept cluster concentration q_m increases. Thus, we define the weight for scaling the uncertainty for for each attribute m as:

$$w_m = 1 - q_m \tag{5}$$

which scales the uncertainty scores for each attribute m based on the dissimilarity of its concept cluster. Finally, the uncertainty estimation for each sample is given by:

$$U^{prompt} = \sum_{m=1}^{M} w_m \frac{U_m^{prompt}}{\log D_m} \tag{6}$$

where the $\log D_m$ term normalizes each entropy $H(p_m) = U_m^{prompt}$ (Eq. 2) so uncertainty scores are comparable across attributes with different numbers of clusters. We then compute the prompt-based uncertainty scores U^{prompt} for all data samples in the data set $\mathcal{I}$. Since we only aim to label a few hundreds samples to fine-tune the LLM, we retain the top-N_s samples from $\mathcal{I}$ to obtain a reduced pool $\mathcal{I}'$ (e.g., $N_s = 1000$) , lowering the computational cost for concept inference and uncertainty estimation in the next phase. Within this current reduced pool $\mathcal{I}'$ from Phase 1, we select top-B_1 samples (e.g., $B_1 = 100$) from $\mathcal{I}'$ for concept labeling and initiating LoRA fine-tuning.

Phase 2: LoRA-Dropout Uncertainty. We adopt Monte Carlo (MC) dropout to quantify the uncertainty of LLM during finetuning with LoRA [12]. The key motivation is that, given limited fine-tuning data (e.g., a few hundred samples), the LoRA adapter can easily overfit to frequent patterns, emitting overconfident predictions that are insensitive to the in-context prompts. In other words, when certain mappings appears many times in the training data, the model increasingly trusts the learned in-weight information over the in-context examples [33]. As a result, forward passes with varying in-context prompts may yield highly similar concept values for unseen samples, collapsing prompt-ensemble uncertainty and obscuring genuine epistemic uncertainty. Similarly to prompt-ensemble uncertainty above, this method is also performed with the following **five steps**: 1) querying the LLM to obtain multiple concept values per attribute; 2) clustering the concept values into semantically coherent groups; 3) computing the variability of concept clusters as uncertainty; 4) weighting the uncertainty of attributes based on inter-cluster similarity; and finally 5) selecting a small set of samples for human-in-the-loop labeling and continued LoRA fine-tuning.

To enable efficient uncertainty estimation, we activate dropout only on the LoRA parameters, and perform K stochastic forward passes with different dropout masks for each input $\boldsymbol{x}$, keeping the task instruction and attribute template fixed in prompt $\boldsymbol{q}$. The K forward passes yield a set of concept values $\{v_m^k\}_{k=1}^K$ for each attribute m, which is then grouped into D_m semantically coherent groups $\{\mathcal{C}_{m,d}\}_{d=1}^{D_m}$ using NLI-based clustering. Following previous work [7], we define the LoRA-dropout epistemic uncertainty for each attribute m as follows:

$$U_m^{\text{dropout}} = H(\bar{p}_m) - \mathbb{E}\left[H(p_m^k)\right], \tag{7}$$

where $H(\bar{p}_m)$ denotes the entropy of the mean probability distribution over concept clusters $\{\mathcal{C}_{m,d}\}_{d=1}^{D_m}$, and $\mathbb{E}\left[H(p_m^k)\right]$ indicates the *expected* entropy of the concept cluster distribution over K forward passes with different dropout masks. Here, it is important to note that for dropout-based uncertainty, $\mathbb{E}\left[H(p_m^k)\right]$ captures the inherent uncertainty due to parameter perturbations, thus subtracting it from $H(\bar{p}_m)$ is necessary to isolate the model's lack of knowledge about the data (epistemic uncertainty). In contrast, there is no parameter perturbation for prompt-ensemble uncertainty, so epistemic uncertainty can be estimated with $H(\bar{p}_m)$ alone. Additionally, while $H(\bar{p}_m)$ can be calculated using Eq. 2, it is difficult to compute $H(p_m^k)$ because this would require re-scoring every cluster under the LLM for each k-th forward pass. Specifically, to obtain $p_m^k(\mathcal{C}_{m,d})$, we need to perform a full cluster-likelihood evaluation per cluster $\mathcal{C}_{m,d}$, per forward pass k, and per attribute m, which is computationally expensive due to the curse of dimensionality, especially for parameter-heavy LLMs.

Therefore, we propose to construct a soft cluster distribution for each k-th forward pass utilizing the NLI-based clustering framework. First, for each attribute m, we extract the concept cluster prototypes $\{\mu_{m,d}\}_{d=1}^{D_m}$ for its concept clusters $\{\mathcal{C}_{m,d}\}_{d=1}^{D_m}$. Then, for each concept value v_m^k generated for attribute m under the k-th forward pass, we score its semantic match to each cluster prototype $\mu_{m,d}$ using bi-directional NLI entailment probabilities as follows:

$$r(v_m^k, \mathcal{C}_{m,d}) = \frac{1}{2}\left(g_{\text{NLI}}(v_m^k \Rightarrow \mu_{m,d}) + g_{\text{NLI}}(\mu_{m,d} \Rightarrow v_m^k)\right), \tag{8}$$

where $g_{\text{NLI}}(v_m^k \Rightarrow \mu_{m,d})$ indicates the probability that concept value v_m^k entails cluster prototype $\mu_{m,d}$ of cluster

$\mathcal{C}_{m,d}$. Consequently, we perform sum-normalization over the concept-cluster scores to form a categorical distribution for each pass k and attribute m, which is given by:

$$p_m^k(\mathcal{C}_{m,d}) = \frac{r(v_m^k, \mathcal{C}_{m,d}) + \varepsilon}{\sum_{l=1}^{D_m}(r(v_m^k, \mathcal{C}_{m,l}) + \varepsilon)}, \tag{9}$$

where $\varepsilon > 0$ is a tiny floor value for numerical stability. Subsequently, we average the concept cluster distribution over K forward passes to obtain the overall concept cluster distribution $\bar{p}_m = \frac{1}{K}\sum_{k=1}^{K} p_m^k(\mathcal{C}_{m,d})$, which results in the formalization for each entropy terms in Eq. 7 as follows:

$$H(\bar{p}_m) = -\sum_{d=1}^{D_m} \bar{p}_m(\mathcal{C}_{m,d}) \log(\bar{p}_m(\mathcal{C}_{m,d})) \tag{10}$$

$$\mathbb{E}\left[H(p_m^k)\right] = \frac{1}{K}\sum_{k=1}^{K}\left[-\sum_{d=1}^{D_m} p_m^k(\mathcal{C}_{m,d}) \log(p_m^k(\mathcal{C}_{m,d}))\right]. \tag{11}$$

Finally, we apply the distance-aware scaling weight w_m (defined in Eq. 5) and division by $\log(D_m)$ normalization described earlier to U_m^{dropout} to obtain the LoRA dropout uncertainty for each data sample as:

$$U^{dropout} = \sum_{m=1}^{M} w_m \frac{U_m^{dropout}}{\log D_m}. \tag{12}$$

Regarding sample selection, we rank the unlabeled samples in the previously mentioned reduced data pool $\mathcal{I}'$ using the uncertainty scores U^{dropout}, and select the top-B_2 samples for human-in-the-loop labeling and continued LoRA fine-tuning in Phase 2.

Algorithm. We also summarize the ConceptAlign algorithm in Appendix 1.

5. Evaluation

In this section, we first evaluate the effectiveness of the proposed ConceptAlign on two image recognition benchmarks. Then, we conduct case studies to validate its efficacy for accurate concept generation and interpretability. In addition, we will perform ablation studies to investigate our two uncertainty strategies on model performance.

5.1. Experimental Settings

Datasets. We evaluate the performance of our method on two benchmark datasets: CUB-200 [29] and Osteoarthritis Initiative (OAI) [9]. The CUB-200 dataset contains 11,780 images with 200 categories of birds. We prompt GPT-4o [2] to enumerate a broad pool of visual attributes for "birds", then manually select 18 attributes characterizing size, shape, color, and marking across multiple bird body parts to form our attribute-concept template. Regarding the OAI dataset, it includes clinical evaluation data, radiological (X-ray and magnetic resonance) images, and a biospecimen repository from 2,877 participants enrolled between February 2004 and May 2006. Following Pierson et al. [21], we collect 21,321 bilateral knee radiographs for joint assessment. In this dataset, domain experts have graded $\{0 : \text{none/doubtful}, 1 : \text{mild}, 2 : \text{moderate}, 3 : \text{severe}\}$ for 10 osteoarthritis (OA) indicators such as *osteophytes*, *subchondral sclerosis*, and *joint-space narrowing (JSN)* across compartments (medial/lateral) and bones (femur/tibia). We use these 10 OA indicators as the high-level attribute template, whose concepts are the expert-assigned scores described above.

Baselines. We will compare the performance of our method with the following competitive baselines: (i) LF-CBM [19], which is the first framework to train CBMs using LLM-generated concepts from class labels instead of human annotations; (ii) VLG-CBM [25] which employs grounded object detection to filter irrelevant LLM-generated concepts for improved interpretability; and (iii) HybridCBM [18], which boosts CBM performance by quantifying the misalignment between between CLIP embeddings of LLM-generated concepts and image features, then translating the missing image representation to augment the concept set. Importantly, all three baselines derive concepts from class labels, in contrast to our approach that learns concepts directly from images with minimal expert supervision.

Implementation Details. We integrate two popular multimodal LLMs, Qwen2.5-VL-7B-Instruct [3] and Llama-3.2-11B-Vision-Instruct [5], to our proposed LLM-based active learning framework. For both models, we apply LoRA to the final 6 Transformer blocks of the vision and language backbones and to the multimodal projector, which mitigates overfitting under limited fine-tuning data and enables effective adaptation across different domains (birds vs. medical). Besides, we optimize the LLM with a token-wise negative log-likelihood (NLL) loss over the target attribute-concept template. The experimental settings and model parameters are detailed in Appendix C.

Prompt Template. During our two-phase pipeline, samples selected from both datasets are driven by dataset-specific *prompt templates*. CUB uses a fixed schema with 18 attributes while OAI uses a fixed schema with 10 attributes. We include the exact prompts, attribute dictionaries, and example JSON outputs in Appendix E.

Metrics. We report accuracy, precision, recall and macro F-1 scores for our experimental results.

5.2. Main Results

In this subsection, we assess the classification performance of ConceptAlign using two image recognition benchmarks. We compare our method with the baselines above. Table 1 summarizes the comparison results for different

Table 1. Comparison of our method and different CBM models on CUB-200 and OAI datasets. **Bold**: Best results on each dataset. Underline: Second best results on each dataset.

Models	CUB-200				OAI			
	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
Supervised Concept-based Models								
LF-CBM [19]	74.58	75.70	74.64	74.53	74.86	72.61	72.36	72.45
VLG-CBM [25]	76.04	79.31	76.81	76.52	75.84	73.77	73.23	73.46
Hybrid-CBM [18]	78.13	78.58	77.24	77.91	73.20	72.41	61.43	64.66
Weakly-Supervised Concept-based Models								
Ours – Qwen2.5	80.02	**80.63**	**80.15**	**80.04**	79.20	78.00	**76.59**	**77.13**
Ours – LLaMa3	**80.19**	80.58	80.11	79.88	**80.07**	**78.63**	75.44	76.64

concept-based models. Overall, the proposed ConceptAlign framework integrated with Qwen2.5 and LLaMa3 outperforms the fully-supervised counterparts on both benchmark datasets. Here, we highlight that ConceptAlign is weakly-supervised, because it utilizes a minimal amount of labeled data to learn meaningful concept values for visual attributes within images. Specifically, only 200 samples from CUB-200 were manually labeled, and for the medical OAI dataset, 300 labeled samples were acquired from human experts. In contrast, all the other supervised baselines require annotated class labels for the entire training data to generate concepts. Below, we further analyze the prediction performance of ConceptAlign compared to the baselines on each benchmark dataset.

Firstly, we examine the prediction performance of ConceptAlign on the CUB-200 benchmark dataset. As shown in Table 1, our ConceptAlign surpasses all concept-based baselines on CUB-200 for bird classification, achieving 80.02% accuracy and 80.04% F1 for Qwen2.5, and 80.19% accuracy and 79.88% F1 with LLaMa3. The effectiveness of ConceptAlign for bird classification lies in its ability to extract high-quality, fine-grained concept values for bird attributes. These high-quality concepts enable the downstream classifier to effectively map subtle visual differences in bird images to distinguishable concept values, which in turn highlight the critical distinctions between bird species. In contrast, the LF-CBM baseline, which relies solely on pretrained models like GPT [2] for concept extraction from class labels and CLIP [22] for concept confidence scoring, demonstrates the poorest performance. This can be attributed to weak concept signals from the pretrained models, whose training data may not capture fine-grained textual concepts relevant to bird images. While VLG-CBM incorporates a grounded object detector [16] to filter out irrelevant concepts generated by LLMs and improve accuracy, the resulting concepts fail to provide sufficient information for the final prediction. Moreover, the Hybrid-CBM method, while achieving performance closest to ours, struggles to learn high-quality concepts from the input data, which are often too high-level or irrelevant, as evident in Figure 1.

Next, regarding the OAI medical dataset, ConceptAlign also outperforms all baselines in the task of knee osteoarthritis (OA) classification. Compared to the CUB-200 dataset, we can observe from Figure 1 an increased accuracy margin of 5% in the classification accuracy between our methods and the baselines. This gain comes from learning domain-specific concepts such as osteophytes, joint-space narrowing, and sclerosis directly from image evidence. Essentially, ConceptAlign allows LLMs to mirror typical clinical workflow, which first extracts concept-level findings from unlabeled data and then aggregates these findings to reach the final diagnosis. On the other hand, other baselines use LLMs to generate medical concepts, where the lack of domain-specific knowledge clearly undermines the prediction performance. Hybrid-CBM performs the worst among all models on the OAI dataset, as shown in Table 1. Upon closer inspection, we find its proposed concept translator [18] generates completely wrong concepts like "muscle", "eye", or "elbow", despite the input X-ray images clearly depicting only the knee. Notably, ConceptAlign requires only 300 labeled samples with concept values annotated by human experts, highlighting its practicality in data-intensive healthcare applications.

5.3. Case Studies

We perform case studies on the CUB-200 dataset to validate the effectiveness of ConceptAlign for generating accurate fine-grained concepts to explain bird classifications.

5.3.1. Do Top-10 Concepts Explain the Model Decision?

We first present a case study to evaluate the quality of bird concepts generated by our proposed ConceptAlign, and assess whether these concepts serve as strong evidence for accurate species identification. As previously mentioned, our downstream classification pipeline adopts a concept bottleneck layer (CBL) to assign numerical activations to the generated concepts, followed by a simple linear predictor over those activations. Thus, following previous work [19], we calculate the contribution of each concept to the final decision with a linear logit decomposition: we first apply *z-score* normalization to the CBL activations across the

dataset and then scale each normalized activation by its learned weight in the linear layer to rank its influence on the final decision.

Figure 4 shows the Top-10 concept contributions from ConceptAlign–LLaMA3 on three example images from the CUB-200 dataset, revealing that fine-grained color and marking concepts localized to specific body parts (e.g., face, chest, and wings) provide strong evidence for the predicted bird species. Specifically in Figure 4, the Florida Jay in sample ② can be identifed by its blue coloration across multiple body parts and an "extremely long tail". Besides, while both birds in sample ① and ③ have black wings and white belly, our method highlights the top distinguishing cues: a "constrasting red-white face" for the European Gold Finch (①) and "red chest patches" for the Rose Breasted Grosbeak (③). Notably, while all baselines recover the "red chest" cue for the Grosbeak (③) from class labels, none of them detect the red facial coloring of the Goldfinch (①), as shown in Figure 1. This result underscores ConceptAlign's superior ability to deliver concept-based explanations for its predictions: without access to extensive class labels, it still effectively learns high-quality concepts directly from the input data.

5.3.2. Are generated concepts faithful to each image?

To assess concept accuracy on CUB-200, we randomly sample 100 test images and manually annotate values for the 18 high-level attributes. For each image–attribute pair, we compare the LLM-generated value with the human annotation using an NLI-based semantic equivalence check [14], counting a match as 1 and a mismatch as 0. A sample's concept accuracy is the mean of its 18 binary matches; the overall concept accuracy is the average across the 100 images. We obtain concept accuracy scores of 75.11% and 79.28% respectively for ConceptAlign–Qwen2.5 and ConceptAlign–LLaMA3. These results highlight that our method can generate accurate concepts that align well with human-annotated ground truths while using minimal labeled data. Additionally, we compare the concept accuracy of ConceptAlign with the baselines in Appendix D.

5.4. Ablation Study and Sensitivity Analysis

We investigate the effect of the two uncertainty strategies on the performance of the proposed method using the OAI dataset. As shown in Table 2, the model performance decreases by approximately 2–2.5% when either uncertainty method is omitted. This result highlights the effectiveness of the proposed hybrid uncertainty approach.

Table 2. Concept accuracy of our two uncertainty methods.

w/o Prompt-Ensemble	w/o LoRA-Dropout	Ours
77.35	76.71	**79.22**

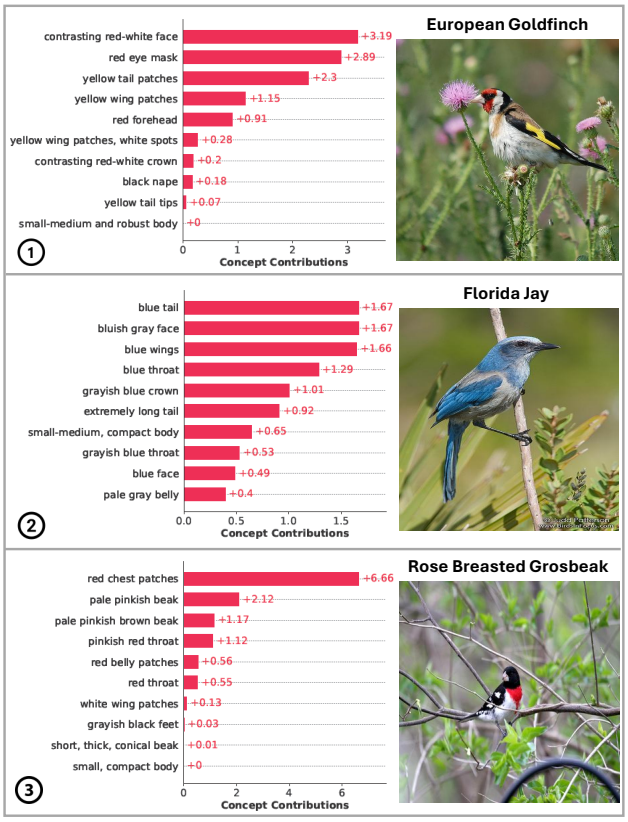

Figure 4. Concept contributions using ConceptAlign-LLaMA3 model for predicting bird species on three samples from the CUB-200 dataset. From the figure, our method can reliably surface a wide range of fine-grained concepts for colors and markings across different bird body parts. Additionally, it elevates these patterns as the primary concepts for species identification.

Effect of important hyperparameters. We also study the impact of important hyperparameters on model performance. The detailed results are introduced in Appendix F.

6. Conclusion

In this work, we developed a novel human-in-the-loop LLM framework for concept generation and alignment via active learning. To identify the most informative data-attribute pairs, we proposed a hybrid uncertainty estimation technique that integrated a prompt-ensemble uncertainty with LoRA dropout. The key advantage of the proposed method lay in its ability to reduce the number of required labeled concepts and data samples. Experiments on two benchmark datasets demonstrated that our approach significantly outperformed the baselines, achieving higher concept accuracy and downstream classification accuracy with only a small amount of labeled data.

References

[1] Yasin Abbasi Yadkori, Ilja Kuzborskij, András György, and Csaba Szepesvari. To believe or not to believe your llm: Iterative prompting for estimating epistemic uncertainty. *Advances in Neural Information Processing Systems*, 37: 58077–58117, 2024. 3

[2] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023. 6, 7

[3] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2. 5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025. 6

[4] Emanuele Buchicchio and Paolo Carbone. Quantifying uncertainty in large language models using monte carlo dropout. *Authorea Preprints*, 2025. 3

[5] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv e-prints*, pages arXiv–2407, 2024. 6

[6] Viet Duong, Qiong Wu, Zhengyi Zhou, Hongjue Zhao, Chenxiang Luo, Eric Zavesky, Huaxiu Yao, and Huajie Shao. Cat: Interpretable concept-based taylor additive models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 723–734, 2024. 1

[7] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *international conference on machine learning*, pages 1050–1059. PMLR, 2016. 5

[8] Amirata Ghorbani, James Wexler, James Y Zou, and Been Kim. Towards automatic concept-based explanations. *Advances in neural information processing systems*, 32, 2019. 1

[9] Tracy Hampton. Osteoarthritis initiative. *JAMA*, 291(16): 1951–1951, 2004. 6

[10] Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. Deberta: Decoding-enhanced bert with disentangled attention. *arXiv preprint arXiv:2006.03654*, 2020. 3

[11] Bairu Hou, Yujian Liu, Kaizhi Qian, Jacob Andreas, Shiyu Chang, and Yang Zhang. Decomposing uncertainty for large language models through input clarification ensembling. In *Proceedings of the 41st International Conference on Machine Learning*, pages 19023–19042, 2024. 3

[12] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022. 1, 5

[13] Pang Wei Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. Concept bottleneck models. In *International conference on machine learning*, pages 5338–5348. PMLR, 2020. 1, 2

[14] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. *arXiv preprint arXiv:2302.09664*, 2023. 3, 8

[15] Akshay Kulkarni, Ge Yan, Chung-En Sun, Tuomas Oikarinen, and Tsui-Wei Weng. Interpretable generative models through post-hoc concept bottlenecks. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 8162–8171, 2025. 1

[16] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, et al. Grounding dino: Marrying dino with grounded pre-training for open-set object detection. In *European conference on computer vision*, pages 38–55. Springer, 2024. 7

[17] Xiaoou Liu, Tiejin Chen, Longchao Da, Chacha Chen, Zhen Lin, and Hua Wei. Uncertainty quantification and confidence calibration in large language models: A survey. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*, pages 6107–6117, 2025. 3

[18] Yang Liu, Tianwei Zhang, and Shi Gu. Hybrid concept bottleneck models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 20179–20189, 2025. 2, 6, 7, 1

[19] Tuomas Oikarinen, Subhro Das, Lam M Nguyen, and Tsui-Wei Weng. Label-free concept bottleneck models. *arXiv preprint arXiv:2304.06129*, 2023. 1, 6, 7, 2

[20] Konstantinos P Panousis, Dino Ienco, and Diego Marcos. Coarse-to-fine concept bottleneck models. *Advances in Neural Information Processing Systems*, 37:105171–105199, 2024. 2

[21] Emma Pierson, David M Cutler, Jure Leskovec, Sendhil Mullainathan, and Ziad Obermeyer. An algorithmic approach to reducing unexplained pain disparities in underserved populations. *Nature Medicine*, 27(1):136–140, 2021. 6

[22] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmLR, 2021. 7, 1

[23] C Radhakrishna Rao. Diversity: Its measurement, decomposition, apportionment and analysis. *Sankhyā: The Indian Journal of Statistics, Series A*, pages 1–22, 1982. 5

[24] Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084*, 2019. 5

[25] Divyansh Srivastava, Ge Yan, and Lily Weng. Vlg-cbm: Training concept bottleneck models with vision-language guidance. *Advances in Neural Information Processing Systems*, 37:79057–79094, 2024. 6, 7, 1, 2

[26] Chung-En Sun, Tuomas Oikarinen, Berk Ustun, and Tsui-Wei Weng. Concept bottleneck large language models. *arXiv preprint arXiv:2412.07992*, 2024. 1, 2

[27] Zhen Tan, Lu Cheng, Song Wang, Bo Yuan, Jundong Li, and Huan Liu. Interpreting pretrained language models via concept bottlenecks. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, pages 56–74. Springer, 2024. 2

[28] Alaa Tharwat and Wolfram Schenck. A survey on active learning: State-of-the-art, practical challenges and research directions. *Mathematics*, 11(4):820, 2023. 3

[29] C. Wah, S. Branson, P. Welinder, P. Perona, and S. Belongie. Caltech-ucsd birds-200-2011. Technical Report CNS-TR-2011-001, California Institute of Technology, 2011. 6

[30] Bowen Wang, Liangzhi Li, Yuta Nakashima, and Hajime Nagahara. Learning bottleneck concepts in image classification. In *Proceedings of the ieee/cvf conference on computer vision and pattern recognition*, pages 10962–10971, 2023. 1, 2

[31] Xinru Wang, Hannah Kim, Sajjadur Rahman, Kushan Mitra, and Zhengjie Miao. Human-llm collaborative annotation through effective verification of llm labels. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, pages 1–21, 2024. 3

[32] Yu Xia, Subhojyoti Mukherjee, Zhouhang Xie, Junda Wu, Xintong Li, Ryan Aponte, Hanjia Lyu, Joe Barrow, Hongjie Chen, Franck Dernoncourt, et al. From selection to generation: A survey of llm-based active learning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14552–14569, 2025. 3

[33] Yasin Abbasi Yadkori, Ilja Kuzborskij, András György, and Csaba Szepesvári. To believe or not to believe your llm. *arXiv preprint arXiv:2406.02543*, 2024. 5

[34] An Yan, Yu Wang, Yiwu Zhong, Zexue He, Petros Karypis, Zihan Wang, Chengyu Dong, Amilcare Gentili, Chun-Nan Hsu, Jingbo Shang, et al. Robust and interpretable medical image classifiers via concept bottleneck models. *arXiv preprint arXiv:2310.03182*, 2023. 2

[35] Yue Yang, Artemis Panagopoulou, Shenghao Zhou, Daniel Jin, Chris Callison-Burch, and Mark Yatskar. Language in a bottle: Language model guided concept bottlenecks for interpretable image classification. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 19187–19197, 2023. 1, 2

[36] Bo Yuan, Yulin Chen, Yin Zhang, and Wei Jiang. Hide and seek in noise labels: Noise-robust collaborative active learning with llms-powered assistance. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10977–11011, 2024. 3

[37] Ze Yu Zhang, Arun Verma, Finale Doshi-Velez, and Bryan Kian Hsiang Low. Understanding the relationship between prompts and response uncertainty in large language models. *arXiv preprint arXiv:2407.14845*, 2024. 3

ConceptAlign: A Human-in-the-Loop Large Language Model Framework for Concept Generation and Alignment

Supplementary Material

A. Notation Summary

In Table 3, we summarize the notations used in the formulation of our proposed hybrid epistemic uncertainty method in Section 4.

B. Active Learning Algorithm

We summarize the proposed two-phase active learning method in Algorithm 1.

Algorithm 1 The Proposed ConceptAlign.

Input: Data pool $\mathcal{I}$, initial labeled set $\mathcal{A}$, number of rounds T, number of forward passes K, number of in-context examples E, phase-1 selection size N_s, phase-1 label budget B_1, phase-2 label budget B_2
Output: Fine-tuned model f, final labeled set $\mathcal{A}$
Initialize unlabeled set: $\mathcal{U} \leftarrow \mathcal{I} \setminus \mathcal{A}$
for round $t = 0$ to T **do**
  if $t = 0$ **then** # Phase 1: Prompt Ensemble
    for $k = 1$ to K **do**
      Sample in-context $\mathcal{E}^k \subset \mathcal{A}$, s.t. $|\mathcal{E}^k| = E$
    end for
    for each sample in $\mathcal{U}$ **do**
      Run K forward passes with $\{\mathcal{E}^k\}_{k=1}^{K}$ as in-context examples
      Compute prompt-based uncertainty in Eq. 6
    end for
    Select top-N_s most uncertain samples $\mathcal{I}' \subset \mathcal{U}$
    Update unlabeled set: $\mathcal{U} \leftarrow \mathcal{I}'$
    Select top-B_1 uncertain samples $\mathcal{S}$ from $\mathcal{U}$
  else # Phase 2: LoRA Dropout
    for each sample in $\mathcal{U}$ **do**
      Run K forwards with LoRA dropout
      Compute LoRA-dropout uncertainty in Eq. 12
    end for
    Select top-B_2 uncertain samples $\mathcal{S}$ from $\mathcal{U}$
  end if
  Acquire human-in-the-loop labels for samples in $\mathcal{S}$
  Update sets: $\mathcal{A} \leftarrow \mathcal{A} \cup \mathcal{S}$, $\mathcal{U} \leftarrow \mathcal{U} \setminus \mathcal{S}$
  Fine-tune model f on labeled data in updated $\mathcal{A}$
end for
return fine-tuned model f, all labeled data in $\mathcal{A}$

C. Model and Hyper-Parameter Settings

Two-phase Active Learning Details. For both active learning phases, we perform $K = 10$ inference passes for each data sample to estimate uncertainty. At the first cold-start phase, each forward pass is conditioned on a distinct set of 3 in-context examples drawn from 10 seed concept-labeled samples. We aggregate predictions to compute prompt-ensemble uncertainty, and use a baseline similarity threshold $s_{\text{base}} = 0.7$ when adjusting cosine similarities between concept clusters for weighting the uncertainty of each attribute. We then retain the top-1000 most uncertain samples to form a reduced unlabeled data pool. From this pool, we select the top 100 uncertain samples for manual labeling on CUB-200, while for OAI 200 concept labels are gathered from available expert annotations. After this step, we insert LoRA adapters ($r = 16, \alpha = 32$ for CUB-200 and $r = 16, \alpha = 64$ for OAI) and finetune the LLM for 20 epochs each round. Additionally, we enable LoRA dropout with fixed rates $p = 0.2$ and $p = 0.3$ for CUB-200 and OAI respectively, and keep dropout active throughout the second phase to induce stochasticity for uncertainty estimation and to regularize fine-tuning. The second active learning phase runs for $T = 5$ rounds. In each round we select an additional 20 samples using LoRA-dropout uncertainty, obtain human-in-the-loop concept labels, finetune the model, then proceed to the next round.

Downstream Classification Details. Following previous works [18, 25], we perform downstream image classification by first encoding each image with a backbone image encoder, then feeding the resulting embedding through a concept bottleneck layer (CBL) that produces numerical activations for the ConceptAlign-generated concepts, and finally applying a linear predictor on these activations to obtain the classification label. To ensure a fair comparison of classification accuracy, for all CUB-200 and OAI experiments, we finetune a ResNet-50 (RN50) backbone on the corresponding dataset and use it as the image encoder for ConceptAlign, LF-CBM [19], and VLG-CBM [25]. Regarding Hybrid-CBM [18], which employs dual encoders to translate missing image features into textual concepts, we instead finetune the image encoder of CLIP-RN50 [22] while keeping the text encoder frozen. Additionally, for both ConceptAlign and all baselines, the concept bottleneck layer and final linear classifier follow the same architecture and training protocol as in previous CBMs [19, 25], with the only difference in dimensionality arising from the number of concepts. This design ensures that any performance differences can be attributed to the quality of the learned concepts rather than variations in the downstream predictor.

Table 3. Summary of notations.

Notation	Definition
$\mathcal{I}$	a set of N observed data samples
M	number of high-level attributes
V_m	number of concept values for attribute m
$\boldsymbol{q}$	input prompt including fixed task instruction and attribute template
K	number of forward inference passes
$\mathcal{E}^k$	set of in-context examples inserted into the prompt of the k-th forward pass during first cold-start active learning phase
v_m^k	concept value generated in the k-th forward pass for attribute m
$\hat{y}_{\text{NLI}}$	label predicted by the NLI model in $\{\text{contradict}, \text{neutral}, \text{entail}\}$
g_{NLI}	entailment probability predicted by the NLI model in $[0, 1]$
$r(v_m^k, \mathcal{C}_{m,d})$	average bi-directional entailment probability between the k-th concept value and the d-th cluster for attribute m
D_m	number of clusters of concept values for attribute m
$\mathcal{C}_{m,d}$	the d-th cluster of concept values for attribute m
$\mu_{m,d}$	cluster prototype of the d-th concept cluster for attribute m
$\boldsymbol{S}$	similarity matrix between D_m cluster prototypes of attribute m
$\boldsymbol{S}^{\boldsymbol{adj}}$	adjusted similarity matrix between D_m cluster prototypes of attribute m
q_m	concept concentration of attribute m
w_m	distance-aware weight of attribute m
p_m	probability distribution over D_m concept clusters for attribute m over K in-context examples
p_m^k	probability distribution over D_m concept clusters for attribute m given the dropout mask of the k-th forward pass
$\bar{p}_m$	mean probability distribution over D_m concept clusters for attribute m over K dropout masks
$H(.)$	entropy function
U_m^{prompt}	prompt-ensemble uncertainty of attribute m
U^{prompt}	prompt-ensemble uncertainty of a data sample averaged over all its attributes
U_m^{dropout}	LoRA-dropout uncertainty of attribute m
U^{dropout}	LoRA-dropout uncertainty of a data sample averaged over all its attributes

D. Baseline Concept Accuracy Comparison

In addition to the case studies in Subsection 5.3, we compare the accuracy of concepts generated by ConceptAlign and the baselines on a random sample of 100 test images from the CUB-200 dataset. For the baselines, which generate concepts from class labels, we manually assess whether each concept is **visually present** in the corresponding image, assigning 1 to a match and 0 to a mismatch. Similar to our evaluation of ConceptAlign-generated concepts, a sample's concept accuracy is the mean of its binary concept matches, and the overall concept accuracy is the average across the 100 images.

Table 4. Concept accuracy of ConceptAlign compared to the baselines on CUB-200. The best result is highlighted in **bold** and the second best is underlined. We can observed that ConceptAlign significantly outperforms the baselines on concept accuracy.

Models	Concept Accuracy
LF-CBM [19]	55.98
VLG-CBM [25]	59.21
Hybrid-CBM [18]	35.17
ConceptAlign - Qwen2.5	<u>75.11</u>
ConceptAlign - LLaMa3	**79.28**

Table 4 summarizes the concept accuracy results of the proposed ConceptAlign models compared to the baselines. ConceptAlign significantly outperforms all baselines on CUB-200, producing concepts that are much more closely aligned with human perception (79.28% with LLaMa3). The reason for this improvement is that our method utilizes structured generation with a template containing visual attributes suitable for describing bird images, enabling the LLM to accordingly inspect the image and generate accurate concept values for those attributes. In contrast, the baselines generate concepts based on textual class labels without such structured and image-grounded prompts,

which often yield concepts that do not match what is actually present in the image. Overall, all baselines achieve under 60% concept accuracy in Table 4. The best baseline VLG-CBM relies on a pretrained grounded object detection model to improve concept accuracy but still struggles because the detector is not specifically designed for fine-grained bird attribute identification. While Hybrid-CBM generates textual concepts from CLIP image features, many of its concepts like "lays 3–7 eggs", "inhabit open areas", or "high, thin trill" are irrelevant to the input image, leading to particularly poor concept accuracy of 35.17%.

E. Concept Generation Prompt Templates

We show the prompts for generating concept values for 18 CUB-200 bird attributes and 10 OAI osteoarthritis attributes in Figure 5 and Figure 6 respectively.

F. Impact of Hyper-Parameters

Effect of Forward Passes for Uncertainty Estimation. We first study the impact of the number of forward passes K in ConceptAlign on concept accuracy using the OAI dataset. Intuitively, increasing K should improve the quality of uncertainty estimates and the selection of LoRA fine-tuning samples. However, we find that the LLM model can easily overfit to limited in-context learning or fine-tuning data during early active learning rounds, which flattens the uncertainty scores when K is large and reduces the contrast between genuinely uncertain and overconfident concept predictions. As shown in Table 5, using a moderate number of forward passes such as $K = 5$ or $K = 10$ achieves high concept prediction accuracy. On the contrary, a larger value $K = 20$ negatively impacts the performance while incurring additional computational cost due to more inference passes.

Table 5. Effect of the number of forward passes for uncertainty estimation on OAI dataset. The best result is highlighted in **bold** and the second best is underlined.

Forward Passes	Concept Accuracy
$K = 5$	**79.39**
$K = 10$	79.22
$K = 20$	77.54

Effect of LoRA Dropout for Uncertainty Estimation. Next, we investigate the effect of utilizing different dropout rates for computing LoRA Dropout uncertainty during the second active learning phase on OAI dataset. Table 6 demonstrates that using a moderate LoRA dropout rate such as $p = 0.3$ or $p = 0.4$ is beneficial to the concept prediction accuracy, since it induces sufficient parameter stochasticity to highlight epistemic uncertainty and guide informative sample selection. On the other hand, too little dropout ($p = 0.2$) produces weak uncertainty signals due to overly similar forward passes, while too much dropout ($p = 0.5$) hurts both uncertainty estimation and fine-tuning due to over-perturbation of the LoRA parameters. Consequently, ConceptAlign performs well when LoRA dropout is set to a moderate value, which balances reliable uncertainty estimation with stable adaptation.

Table 6. Effect of LoRA dropout rates for uncertainty estimation on OAI dataset. The best result is highlighted in **bold** and the second best is underlined.

Dropout Rates	Concept Accuracy
$p = 0.2$	74.71
$p = 0.3$	79.22
$p = 0.4$	**79.38**
$p = 0.5$	74.58

Effect of Baseline Similarity for Uncertainty Weighting. Moreover, we examine the effect of the baseline similarity threshold s_{base} used for adjusting the concept cluster similarity matrix in Eq. 4, and subsequently weighting the uncertainty score for each attribute in Eq. 6 and Eq. 12. As illustrated in Table 7, concept accuracy on the OAI dataset is maximized at $s_{\text{base}} = 0.7$ (79.22%), with $s_{\text{base}} = 0.8$ having nearly identical performance (79.18%). In contrast, smaller values such as $s_{\text{base}} = 0.5$ and $s_{\text{base}} = 0.6$ degrade concept accuracy because they under-penalize moderately dissimilar clusters, and thus under-weight the uncertainty of difficult attributes whose concept clusters still differ semantically. Additionally, an overly large baseline similarity $s_{\text{base}} = 0.9$ ignores most cluster similarity scores and effectively collapse the distance-aware weighting toward an unweighted measure, leading to worse concept accuracy.

Table 7. Effect of baseline similarity for attribute-wise uncertainty weighting on OAI dataset. The best result is highlighted in **bold** and the second best is underlined.

Baseline Similarity	Concept Accuracy
$s_{\text{base}} = 0.5$	76.37
$s_{\text{base}} = 0.6$	78.80
$s_{\text{base}} = 0.7$	**79.22**
$s_{\text{base}} = 0.8$	79.18
$s_{\text{base}} = 0.9$	79.02

Generation Prompt Template for CUB-200 Attributes

[SYSTEM] You are an expert in analyzing bird images. Your task is to examine a bird image and describe its visual attributes.

Base your descriptions only on what is visually observable in the image. Do not use species-specific knowledge, prior information, or external assumptions that are not directly visible.

Please respond using this exact JSON template:

[
{"name" : "body shape", "value": "..."},
{"name" : "crest shape", "value": "..."},
{"name" : "beak shape", "value": "..."},
{"name" : "neck shape", "value": "..."},
{"name" : "wing shape", "value": "..."},
{"name" : "tail shape", "value": "..."},
{"name" : "flight feather shape", "value": "..."},
{"name" : "feet shape", "value": "..."},
{"name" : "crown color and marking", "value": "..."},
{"name" : "beak color and marking", "value": "..."}
{"name" : "eye color and marking", "value": "..."}
{"name" : "face color and marking", "value": "..."}
{"name" : "throat color and marking", "value": "..."}
{"name" : "upper chest color and marking", "value": "..."}
{"name" : "lower belly color and marking", "value": "..."}
{"name" : "upside tail color and marking", "value": "..."}
{"name" : "upside wing color and marking", "value": "..."}
{"name" : "feet color and marking", "value": "..."}
]

Formatting rules:

- Every attribute must have a complete, non-empty value. Do not leave any attribute blank or incomplete.
- If the attribute name includes "shape", the value must follow the format: "[size and shape] [body part]".
  Examples: {"name": "body shape", "value": "small and compact body"}, {"name": "tail shape", "value": "long and forked tail"}.
- If the attribute name includes only "color", the value must follow the format: "[color(s)] [body part]".
  Example: {"name": "feet color", "value": "pink feet"}.
- If the attribute name includes both "color" and "marking", the value must follow the format: "[color(s)] [body part] with [marking(s)]". Example: {"name": "crown color and marking", "value": "yellow crown with black stripe"}. If no markings are visible, omit the "with [marking(s)]" phrase and use the color only format (e.g., {"name": "crown color and marking", "value": "yellow crown"}).
- If the attribute mentions a body part whose shape, color, or markings cannot be confidently described due to occlusion, blur, angle, or lighting, the value must follow the format: "not clearly visible [body part]". Example: {"name": "lower belly color and marking", "value": "not clearly visible belly"}.
- If the attribute name includes specific body part location such as "upper chest" or "upside tail", only describe that specific body part location. For example, if the attribute is "upside tail color and marking", the value should only describe the color and markings of the upside part of the tail, not the underside (e.g., {"name": "upside tail color and marking", "value": "yellow tail with dark tip"}). If only the underside tail is visible, use the format: {"name": "upside tail color and marking", "value": "not clearly visible tail"}.
- Specifically for the attribute "crest shape", if the head is clearly visible and the crest is clearly absent, use the format: {"name": "crest shape", "value": "no crest"}. If the head and crest are not visible, use the format: {"name": "crest shape", "value": "not clearly visible crest"}.

[USER] [IMAGE] Carefully examine the bird in the given image and describe exactly 18 attributes of the bird listed in the JSON template. Follow the JSON structure and formatting rules from the system prompt. Do not add any explanations or extra text outside the JSON template.

[ASSISTANT] · · ·

Figure 5. Generation Prompt Template for CUB-200 Attributes

Generation Prompt Template for OAI Attributes

[SYSTEM] You are an expert radiologist analyzing knee X-ray images. Your task is to assign compartment-specific grades based on the OARSI (Osteoarthritis Research Society International) grading system for osteophytes, sclerosis, and joint space narrowing attributes.

Base your judgments only on what is visually observable in the image. Do NOT use patient history, prior visit notes, or assumptions that are not directly visible.

Knee compartment definitions:
- medial: left side (opposite the fibula)
- lateral: right side (with the fibula)
- femur: upper side of the joint (thigh bone)
- tibia: lower side of the joint (shin bone)

Attribute definitions:
- osteophytes femur medial grade: bone spurs at the left joint edge on the thigh bone.
- sclerosis femur medial grade: whiter, denser bone just under the left joint surface on the thigh bone.
- joint space narrowing medial grade: narrowing of the space between the thigh bone and shin bone on the left side.
- osteophytes tibia medial grade: bone spurs at the left joint edge on the shin bone.
- sclerosis tibia medial grade: whiter, denser bone just under the left joint surface on the shin bone.
- osteophytes femur lateral grade: bone spurs at the right joint edge on the thigh bone.
- sclerosis femur lateral grade: whiter, denser bone just under the right joint surface on the thigh bone.
- joint space narrowing lateral grade: narrowing of the space between the thigh bone and shin bone on the right side.
- osteophytes tibia lateral grade: bone spurs at the right joint edge on the shin bone.
- sclerosis tibia lateral grade: whiter, denser bone just under the right joint surface on the shin bone.

OARSI grading scale:
- “0”: no osteophytes/sclerosis/joint space narrowing.
- “1”: mild osteophytes/sclerosis/joint space narrowing.
- “2”: moderate osteophytes/sclerosis/joint space narrowing.
- “3”: severe osteophytes/sclerosis/joint space narrowing.

Please respond using this exact JSON template:

```
[
 {“name” : “osteophytes femur medial grade”, “value”: “...”},
 {“name” : “sclerosis femur medial grade”, “value”: “...”},
 {“name” : “joint space narrowing medial grade”, “value”: “...”},
 {“name” : “osteophytes tibia medial grade”, “value”: “...”},
 {“name” : “sclerosis tibia medial grade”, “value”: “...”},
 {“name” : “osteophytes femur lateral grade”, “value”: “...”},
 {“name” : “sclerosis femur lateral grade”, “value”: “...”},
 {“name” : “joint space narrowing lateral grade”, “value”: “...”},
 {“name” : “osteophytes tibia lateral grade”, “value”: “...”},
 {“name” : “sclerosis tibia lateral grade”, “value”: “...”}
]
```

Formatting rules:
- Every attribute must have a complete, non-empty value.
- For each attribute, assign the value as a string OARSI grade in {“0”, “1”, “2”, “3”}.
- Examples: {“name”: “joint space narrowing medial grade”, “value”: “2”}, {“name”: “osteophytes femur lateral grade”, “value”: “0”}, etc.
- Do NOT assign any value outside {“0”, “1”, “2”, “3”}.
- Do NOT include any extra texts, units, or explanations outside the JSON template.

[USER] [IMAGE] Carefully examine the knee X-ray in the given image and assign a value for each listed attribute in the JSON template. Each value is a string OARSI grade in {“0”, “1”, “2”, “3”}. Refer to the provided definitions and grading scale in the system prompt during your assessment of the image. Follow the JSON structure and formatting rules from the system prompt.

[ASSISTANT] · · ·

Figure 6. Generation Prompt Template for OAI Attributes